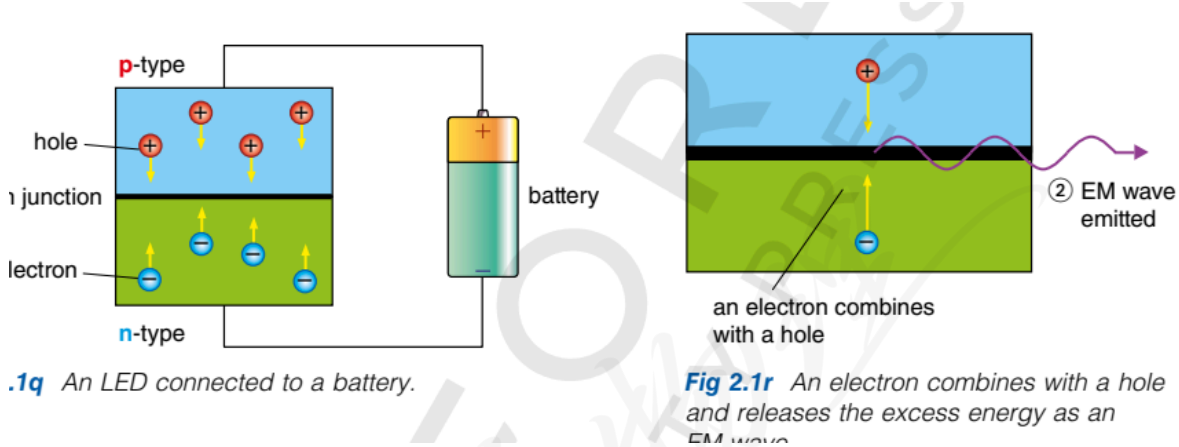
E3.2.1 How LEDs produce light

A hole is a missing electron in a semiconductor and behaves as a mobile positive charge carrier. Electrons are the majority carriers in an n-type region; holes are the majority carriers in a p-type region. Their boundary is a p–n junction.

Under forward d.c. bias, electrons and holes move towards the junction and recombine, releasing energy as electromagnetic radiation. The material's energy difference determines photon energy and emitted colour.



.1q An LED connected to a battery.

Fig 2.1r An electron combines with a hole and releases the excess energy as an EM wave

Forward bias makes electrons and holes recombine at the junction and emit light.

E3.2.2 Luminous flux

Luminous flux weights visible radiant output by the eye's wavelength sensitivity and is measured in lumens. Equal radiant powers with different spectra can have different luminous fluxes. Flux describes total visually effective light output or passage through a surface. Illuminance additionally divides received flux by area and has different units.

E3.2.3 Illuminance

Illuminance is luminous flux received per unit surface area, measured in lux, equivalent to lm m^{-2}. Spreading the same flux over a larger area reduces average illuminance. The area-based calculation gives an average. With non-uniform lighting it does not imply the same illuminance at every point.

$$E = \frac{\Phi}{A}$$

E: illuminance (lx); Φ: luminous flux received by the surface (lm); A: illuminated area (m^2). Uniform illumination, or average illuminance over A.

E3.2.4 Distance, incidence angle and illuminance

Light from an isotropic point source spreads out, so normal-incidence illuminance follows an inverse-square distance law. Oblique incidence spreads light over a larger surface, introducing the cosine of the angle from the normal. An angle measured from the surface needs conversion. Total source flux is not all received by a small target.

$$E = E_{\perp}\cos\theta = \frac{\Phi\cos\theta}{4\pi r^2}$$

Isotropic point source: Φ: total emitted luminous flux (lm); r: source-to-surface distance (m); θ: angle between incident light and surface normal; E: lx.

E3.2.5 Luminous efficacy

Luminous efficacy is luminous flux per watt of electrical input, measured in lm W^{-1}. For the same required flux, a lamp with higher efficacy usually needs less electrical power. This flux-to-power ratio can have a numerical value above 1.

Different lamps convert electrical energy to visible light by different mechanisms, giving different efficacies. Compare the supplied flux, power and lighting conditions.

Lamp	Light-production process	Energy and visible output
Incandescent	Current heats the filament and excites electrons; radiation is emitted as electrons return to lower energy levels.	Much energy becomes heat and infrared; visible output is a small fraction
Fluorescent	Discharge excites mercury atoms to emit ultraviolet; phosphor absorbs it and emits visible light	Converts input energy to visible light more effectively
LED	Electrons and holes recombine at a p–n junction	Can emit visible light directly; material determines colour

$$\eta_v = \frac{\Phi}{P_{\text{in}}}$$

Luminous efficacy η_{v} (lm W^{-1}); Φ: useful luminous flux (lm); P_{in}: input electrical power (W). Not a dimensionless energy efficiency.